Fang-Yi Lin

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Education

National Taiwan University of Science and Technology(NTUST) Taipei, TW
M.S. in Management Information Systems *2023 – 2025*
- Awarded Graduate Excellence Admission Scholarship

National Taipei University(NTPU) New Taipei, TW
B.S. in Statistics *2019-2022*

Experience

Big Data Analysis & Cybersecurity Intern (CAP Program) Feb 2025 – June 2025
Cathay Life Insurance
- Architected a malicious URL classification model using a hybrid **BERT Embedding + CNN** framework, significantly improving prediction accuracy for real-time threat detection.
- Streamlined data processing and training pipelines, achieving a 6x increase in training efficiency through pipeline optimization.
- Developed custom data analysis and visualization toolsets to facilitate feature exploration and iterative model refinement.

Technical Assistant (Industry-Academia Collaboration) 2024
Taipei Fubon Bank
- Facilitated the deployment of the "AI Eagle Eye" anti-fraud model across multiple banking institutions, providing technical consultancy and training to ensure successful production integration.
- Enhanced model interpretability and decision transparency by developing explainability tools based on LIME (Local Interpretable Model-agnostic Explanations).

Projects

Cybersecurity LLM (GAI Focus) | *LangChain, Pytorch, Ollama, Transformer* Assoc. Prof. Raymund Lin
- LLM Optimization: Leveraged Ollama, Prompt Engineering, and LangChain to build a cybersecurity AI assistant with improved contextual understanding and response consistency.
- LLM Fine-tuning: Fine-tuned a cybersecurity-specific language model using Transformers, reducing validation loss from 2.50 to 2.44 and achieving an 11.56 perplexity score, resulting in higher-quality generation.

Financial Trend Prediction & Text Mining | *XGBoost, Text Mining, Embedding* Asst. Prof. Li-Wei Yang
- Processed GB-scale financial text data using TF-IDF for text mining and keyword vectorization to build a robust stock trend prediction model.
- Mitigated data imbalance issues using Oversampling (SMOTE) and utilized PCA for dimensionality reduction, optimizing model performance and computational efficiency.
- Executed 90-day rolling backtests using XGBoost, achieving a 80% prediction accuracy rate.

Awards

AI Postal Innovation - 2024 Big Data Competition | *Technical Lead* Finalist (Top 15 out of 100+ teams)
- Engineered a data-driven ibox site selection and user behavior analysis system, utilizing Machine Learning models to extract key features and predict optimal service locations.
- Architected user behavior models (dispatching/receiving) on AWS, optimizing service distribution strategies based on cloud-scale data insights.
- Developed an interactive visualization platform using Streamlit and Folium, integrating geospatial information to enhance decision-making efficiency and strategy transparency.

Technical Skills

Languages: Python, SQL, R, SAS
AI & Machine Learning: Generative AI, LLM, Model Fine-tuning, BERT, CNN, XGBoost, Prompt Engineering, LIME (Model Explainability)
Frameworks & Libraries: LangChain, Transformers, Scikit-learn, Pandas, NumPy, Pycaret
Developer Tools: Git, GCP, AWS
Data Visualization: Power BI, Tableau, Streamlit